Computer Science & IT



C Programming

Data Types and Operators

Lecture No. 03



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Recap of Previous Lecture



- Topic: Integer
- Topic: Range of unsigned
- Topic: Range of signed Number
- Topic: Character: ASCII value
- Topic: Range of Character

Slide

Topics to be Covered



Topic Problem Solving

Topic Operator & Expression

Topic Associativity & precedence

Topic

Topic

Slide



Topic : Question

Consider the following program

```
#include <stdio.h>
int main( ){
      short int a = 32771;
      printf("%hd", a);
      return 0;
}
```

The output of the above program is -_____

16 bit

0 — 32767

-1 — -32768

65536

32771

-32765

Slide



Topic : Question

Consider the following program

```
#include <stdio.h>
int main( ){
      short int a = 32771;
      printf("%hd", a);
      return 0;
}
```

-32765

The output of the above program is -_____

Slide



Topic : Question

0 — 32767

-1 — -32768

Consider the following program

```
#include <stdio.h>
int main(){
      short int a = 37780;
      printf("%hd", a);
      return 0;
}
```

Negative

-(65536 − 37780)

= 27756

The output of the above program is - _____

Slide



Topic : Question

Consider the following program

```
#include <stdio.h>
int main(){
    short int a = 37780;
    printf("%hd", a);
    return 0;
}
```

$-(65536 - 37780) = -27756$

The output of the above program is - _____

Slide



Topic : Question

Consider the following program

```
#include <stdio.h>
int main(){
    short int a = -33333;
    printf("%hd", a);
    return 0;
}
```

The output of the above program is -

0 — 32767
-1 — -32768
} 16 bits
2^{16}

-33333 → Negative

+ve No (65536 - 33333 = 32203)



Topic : Question

Consider the following program

$(65536 - |-33333|) = 32203$

```
#include <stdio.h>
int main(){
    short int a = -33333;
    printf("%hd", a);
    return 0;
}
```

The output of the above program is -



Topic : Question

0 —— 32767

Consider the following program

```
#include <stdio.h>
int main(){
      short int a = 64501;
      printf("%hd", a);
      return 0;
}
```

Negative

65536
− 64501
1035

−1035

The output of the above program is -

Slide

Short int a = −65540;

Negative

−65540 % 65536

= −4

−1 ——— −32767

7 | Q : 2

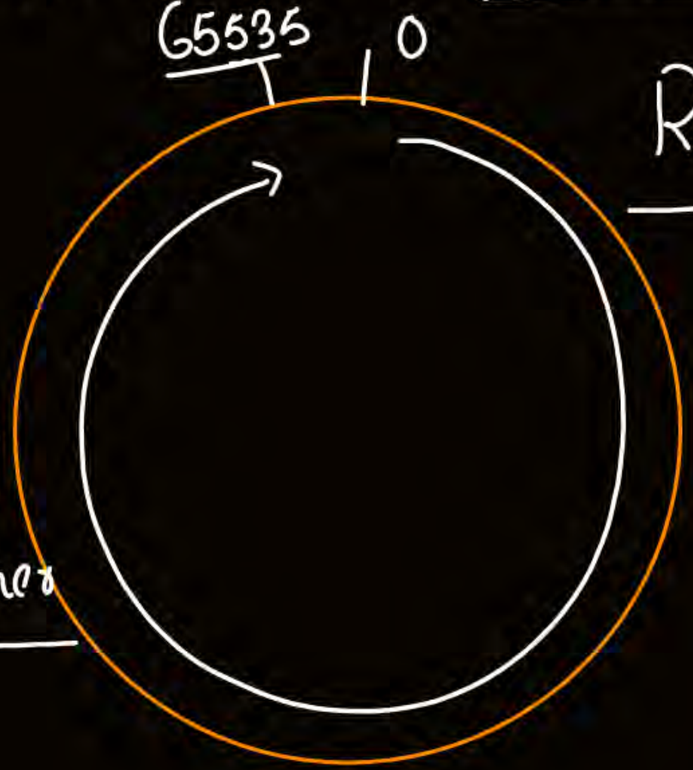
3 | R : 1

Unsigned — 0 - 65535

Short int a = 65540;

0 — 32767

4



Remainder

2^{16}

65540 % 65536

4

7/3

2 - Quotient

1 - Remainder

$0 - 2^{n-1} - 1]$

printf("%hd",



Topic : Question

Consider the following program

$-(65336 - 64501) = -1035$

```
#include <stdio.h>
int main(){
    short int a = 64501;
    printf("%hd", a);
    return 0;
}
```

The output of the above program is -

Slide _ _ _ _ _ _ _ _ _ _ _ _ _ _ _



Sign Character

```
#include <stdio.h>
int main(){
        char a = 128;
        char c = -128;
        printf("%c\n", a);
        printf("%d\n", a);
        printf("%c\n", c);
        printf("%d", c);
        return 0;
}
```



Signed Char

```c
#include <stdio.h>
int main() {
    char ch = -134;
    printf("%c", ch);
    return 0;
}
```

86 bits





$$\frac{(256 - |-134|)}{134}$$

$$122$$

Slide

128	Ç	144	É	160	á	176	▒	192	└	208	╨	224	α	240	≡				
129	ü	145	æ	161	í	177	▒	193	┴	209	╤	225	ß	241	±				
130	é	146	Æ	162	ó	178	▓	194	┬	210	╥	226	Γ	242	≥				
131	â	147	ô	163	ú	179	│	195	├	211	╙	227	π	243	≤				
132	ä	148	ö	164	ñ	180	┤	196	─	212	╘	228	Σ	244	⌠				
133	à	149	ò	165	Ñ	181	╡	197	┼	213	╒	229	σ	245	⌡				
134	å	150	û	166	ª	182	╢	198	╞	214	╓	230	µ	246	÷				
135	ç	151	ù	167	º	183	╖	199	╟	215	╫	231	τ	247	≈				
136	ê	152	ÿ	168	¿	184	╕	200	╚	216	╪	232	Φ	248	°				
137	ë	153	Ö	169	⌐	185	╣	201	╔	217	┘	233	Θ	249	∙				
138	è	154	Ü	170	¬	186	║	202	╩	218	┌	234	Ω	250	·				
139	ï	155	¢	171	½	187	╗	203	╦	219	█	235	δ	251	√				
140	î	156	£	172	¼	188	╝	204	╠	220	▄	236	∞	252	n				
141	ì	157	¥	173	¡	189	╜	205	═	221	▌	237	φ	253	2				
142	Ä	158	₧	174	«	190	╛	206	╬	222	▐	238	ε	254	■				
143	Å	159	ƒ	175	»	191	┐	207	╧	223	▀	239	∩	255					

Source: www.LookupTables.com

www.LookupTables.com

Slide

Dec	Hex	Name	Char	Ctrl-char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	0	Null	NUL	CTRL-@	32	20	Space	64	40	@	96	60	`
1	1	Start of heading	SOH	CTRL-A	33	21	!	65	41	A	97	61	a
2	2	Start of text	STX	CTRL-B	34	22	"	66	42	B	98	62	b
3	3	End of text	ETX	CTRL-C	35	23	#	67	43	C	99	63	c
4	4	End of xmit	EOT	CTRL-D	36	24	$	68	44	D	100	64	d
5	5	Enquiry	ENQ	CTRL-E	37	25	%	69	45	E	101	65	e
6	6	Acknowledge	ACK	CTRL-F	38	26	&	70	46	F	102	66	f
7	7	Bell	BEL	CTRL-G	39	27	'	71	47	G	103	67	g
8	8	Backspace	BS	CTRL-H	40	28	(	72	48	H	104	68	h
9	9	Horizontal tab	HT	CTRL-I	41	29	)	73	49	I	105	69	i
10	0A	Line feed	LF	CTRL-J	42	2A	*	74	4A	J	106	6A	j
11	0B	Vertical tab	VT	CTRL-K	43	2B	+	75	4B	K	107	6B	k
12	0C	Form feed	FF	CTRL-L	44	2C	,	76	4C	L	108	6C	l
13	0D	Carriage feed	CR	CTRL-M	45	2D	-	77	4D	M	109	6D	m
14	0E	Shift out	SO	CTRL-N	46	2E	.	78	4E	N	110	6E	n
15	0F	Shift in	SI	CTRL-O	47	2F	/	79	4F	O	111	6F	o
16	10	Data line escape	DLE	CTRL-P	48	30	0	80	50	P	112	70	p
17	11	Device control 1	DC1	CTRL-Q	49	31	1	81	51	Q	113	71	q
18	12	Device control 2	DC2	CTRL-R	50	32	2	82	52	R	114	72	r
19	13	Device control 3	DC3	CTRL-S	51	33	3	83	53	S	115	73	s
20	14	Device control 4	DC4	CTRL-T	52	34	4	84	54	T	116	74	t
21	15	Neg acknowledge	NAK	CTRL-U	53	35	5	85	55	U	117	75	u
22	16	Synchronous idle	SYN	CTRL-V	54	36	6	86	56	V	118	76	v
23	17	End of xmit block	ETB	CTRL-W	55	37	7	87	57	W	119	77	w
24	18	Cancel	CAN	CTRL-X	56	38	8	88	58	X	120	78	x
25	19	End of medium	EM	CTRL-Y	57	39	9	89	59	Y	121	79	y
26	1A	Substitute	SUB	CTRL-Z	58	3A	:	90	5A	Z	122	7A	z
27	1B	Escape	ESC	CTRL-[	59	3B	;	91	5B	[	123	7B	{
28	1C	File separator	FS	CTRL-\	60	3C	<	92	5C	\	124	7C	|
29	1D	Group separator	GS	CTRL-]	61	3D	=	93	5D	]	125	7D	}
30	1E	Record separator	RS	CTRL-^	62	3E	>	94	5E	^	126	7E	~
31	1F	Unit separator	US	CTRL-_	63	3F	?	95	5F	_	127	7F	DEL

Slide



UnSigned value

(d)
Unsigned value for –134= 256 – 134 = 122.
Hence, ‘z’ is printed.



Signed Char

```
#include <stdio.h>
int main() {
char ch = -139;
printf("%c", ch);
return 0;
}
```

The output is-

(a) u

(b) Non printable output

(c) Overflow warning

(d) U



0 — 127

-1 — -128

256

139

117

a - 97

122 z

121 y

120 x

119 w

118 v

117 u



topic : variable

Consider the following program:

```
#include <stdio.h>
int main(){
char ch=112;
ch=ch+6;
printf("%c", ch);
return 0; }
```

The output is _____.

Slide



topic : variable

Consider the following program:

```
#include <stdio.h>
int main(){
char ch=112;
ch=ch+6;
printf("%c", ch);
return 0; }
```

The output is _____.

v is the output

Slide



Topic : Variable name

identifier

int a; variable

Rule : ~~a/p~~ alphabet (a..z, A..z)

Number (0 — 9)

underscore

Start will be

_ or alphabet

↑ underscore

Slide



Topic : Variable name

Variable name must start with _ (underscore) or alphabet. length- 31 character.

Slide



Topic : Reserve keyswords

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while

Reserved keywords can't be the name of variable



Topic : Reserve keyswords

Which of the following is not a valid C variable name ?

$ ← Special symbol

A `int number;`

B `int $ main;`

C `int variable_count;`

D `float rate;`

Slide



Topic : Reserve keyswords

```c
#include <stdio.h>
int main(){
char Int=112;
Int=Int+6;
printf("%c", Int);
return 0;
}
```

Correct

Int

int

Int

C - language

Case Sensitive



Topic : Reserve keyswords

```
#include <stdio.h>
int main(){
char _ch=112;
_ch=_ch+6;
printf("%c", _ch);
return 0;
}
```

is this valid character name

alphabet / _ (underscore)

`_' only underscore.



Topic : Operators

```
#include <stdio.h>
int main(){
char _ch=112;
_ch=_ch+6;
printf("%c", _ch);
return 0;
}
```



Topic : User Input

scanf function

```
# include <stdio.h>
int a;
printf("Enter the Number:")
Scanf("%d", &a);
```

a | 20 | 1000

& address of

operator &

```
Enter the Number:
20
```



Topic : User Input

scanf fuction

```
#include <stdio.h>
int main(){
        int a;
        printf("Ennter the value of a :");
        scanf("%d", &a);
        printf("Ennter the value of a %d:", a);
}
```



Topic : User Input

scanf fuction

```
#include <stdio.h>int main(){
        int a, b;
        printf("Ennter the value of a and b :");
        scanf("%d %d", &a, &b);
        printf("Ennter th value of a %d and b is %d:", a, b);
}
```

& - address of

pointer

Slide

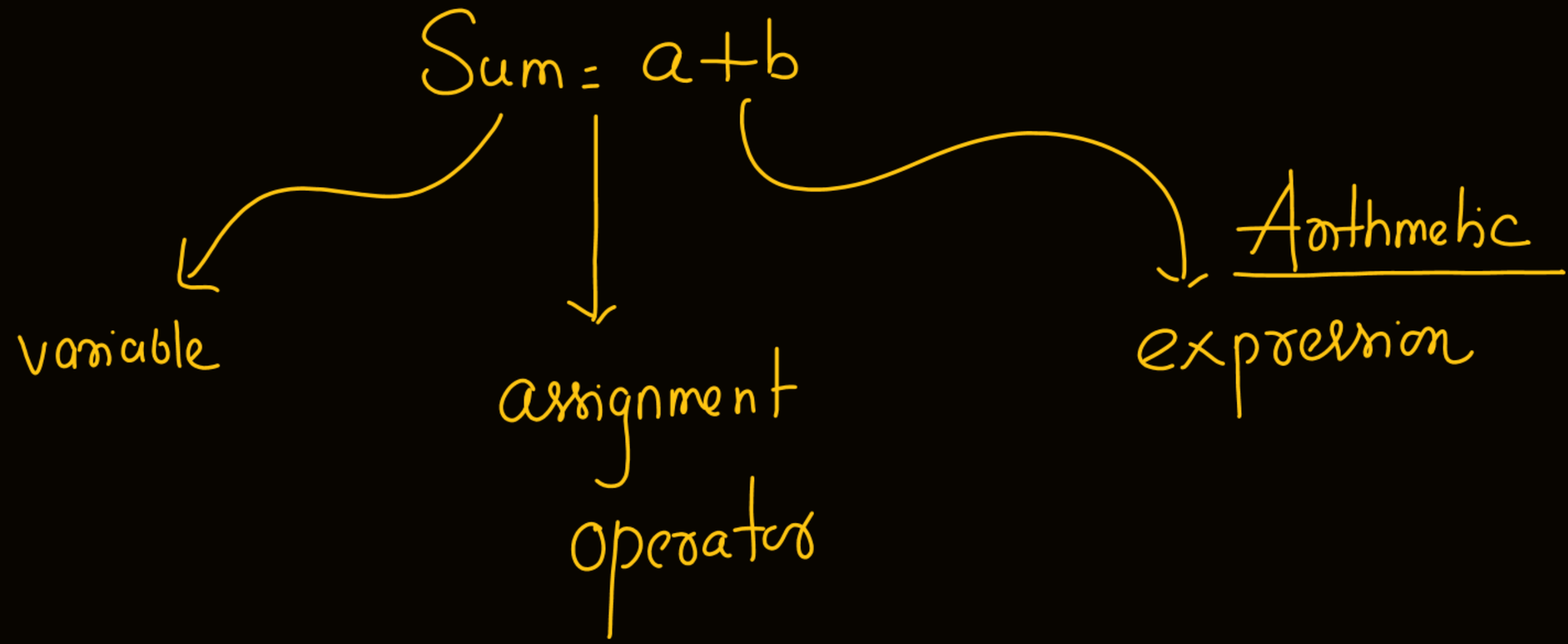


Topic : Question

Write a program to add 2 numbers

```
int main() {
    int a=20, b=10;
    int Sum;
    Sum = a+b;
    printf("Sum of a and b is %d", Sum);
    return 0;
}
```

Slide


Sum = a+b
variable
assignment operator
Arithmetic expression



Topic : Operators

Operators :

- Unary : operand
- Binary : operand
- Ternary : ? Conditional

a + b
- a, b → operand
- + → operator

Scanf("%d", &a) → unary operator

int a = -5 → unary operator

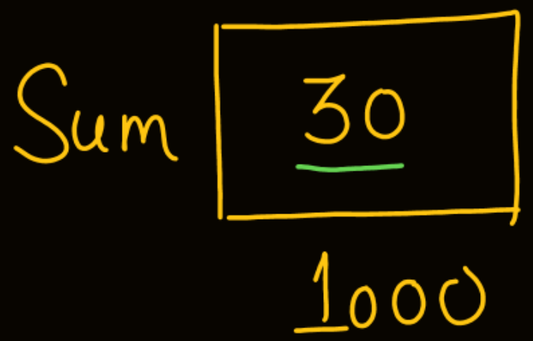
Sum = a + b → binary operator

of operand. 2

Slide

Assignment operator : Sum = a+b

L-value & R-value



Sum = a + b

Sum L-value address associated with Sum (1000)

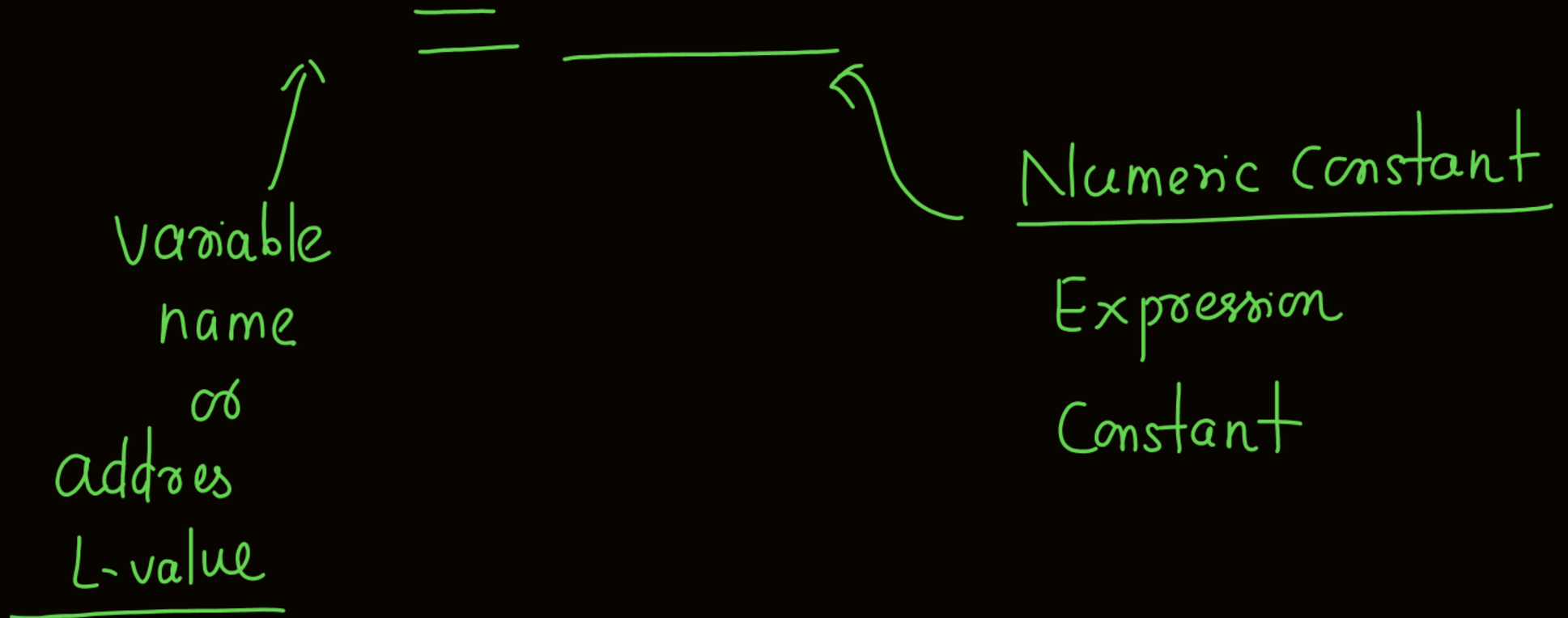
Sum R-value is value store in location of sum. (30)

In assignment operation R-value of right Side of assigment store in L-value of Left Side of assigment

(30)

Sum = a + b

(1000)

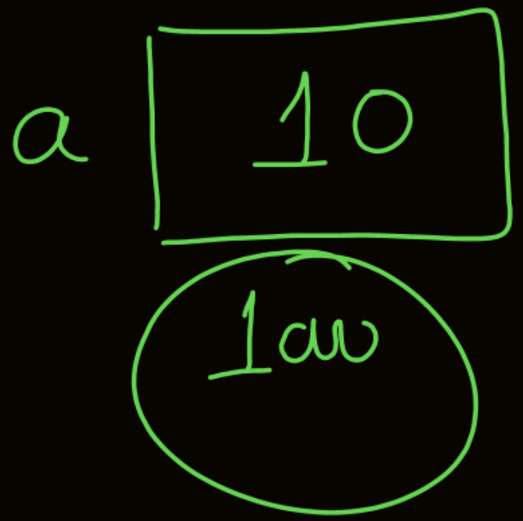

=
Variable name or addres L-value
Numeric Constant
Expression
Constant

&a;

int a = 10;

↑ constant

Numeric



printf("%u", &a);

? (3 operand)

Scanf("%d", &a)

a | 2000 | 1000

↑ user input



Topic : Operators

Assignment Operators

```
#include <stdio.h>
      int main() {
            int a = -129;
            a = 120
            printf("%d",a);
            return 0;
      }
```

t.me/Abhishekshaarmapw



2 mins Summary

&pic

&pic

&pic

&pic

&pic

Slide

THANK - YOU

Slide